MODULE 04

Memory

The Secret That Nobody Teaches

Structured memory architecture: how to make your agent remember, learn, and evolve between sessions

DURATION
25 min

FORMAT
Demo + Diagrams

KIT
PRD + 7 Templates

The Problem: Digital Alzheimer's

Every time a session ends, your agent forgets everything. Decisions you made yesterday? Forgot. Preferences you explained 3 times? Forgot. Ongoing projects? Forgot.

Without structured memory, you repeat context every single day. It's like hiring an assistant brilliant that wakes up with amnesia every morning.

··Real case · Day 2

Session exceeded 173k+ tokens (limit 160k). Agent stopped responding. Cause: zero compaction configured, context growing infinitely. Fix: configure compaction + reserveTokensFloor BEFORE needing it.

The Solution: Layered Memory

Instead of relying only on session context, we created a 4-layer architecture · from ephemeral to permanent:

Session
Current context
·
Daily Note
memory/YYYY-MM-DD.md
·
Topic Files
decisions, lessons...
·
MEMORY.md
Central index

Each layer filters and distills. The session is raw. The daily note is record. The topic files are curation. The MEMORY.md is distilled wisdom.

Specialized Topic Files

Inside memory/ , each file has a clear purpose:

decisions.md · Permanent Decisions

Everything that has been decided and should NOT be reversed without discussion. "All credentials in 1Password", "Sonnet for crons", "Hub model > Mesh model". Never lose.

lessons.md · Lessons Learned

Errors, patterns, insights. With intelligent retention: · Strategic = permanent (patterns, philosophy). · Tactics = expire in 30 days (bugs, workarounds). Monthly review.

projects.md · Active Projects

Current state of each project: what's in progress, what's stopped, what's completed. The agent consults before suggesting work.

people.md · People & Team

Contacts, partners, team. Who does what, how to communicate, preferences of each person. Essential for multi-agents.

pending.md · Awaiting Input

Everything that depends on you (human). The agent doesn't forget · and holds you accountable. "Bruno needs to provide Anthropic API key", "Awaiting decision on pricing".

MEMORY.md as Index

The MEMORY.md at the root of the workspace is the index · points to the topic files without duplicating content. It is the "executive summary" of the agent's memory. Loaded at each session.

MODULE 04 · MEMORY
Build Your AI COO
Compaction & The Inviolable Rule
Configuring Compaction
Without compaction, your session grows until it bursts. The essential config:
·· Compaction Config
```
{
  "compaction": { "mode": "default" },
  "contextTokens": 160000,
  "reserveTokensFloor": 30000
}
```
The reserveTokensFloor ensures that the agent finishes reasoning before compacting · without it, answers get cut off in the middle.
·INVIOLABLE RULE
Before EACH compaction, the agent MUST extract: lessons · lessons.md, decisions · decisions.md, pending items · pending.md. If it doesn't extract before compacting, it loses 80% of the value
. It's like formatting the HD without a backup.
Feedback Loops: Approve/Reject
The feedback system makes the agent learn from its decisions. When you reject a suggestion, the reason is noted · and consulted before the next suggestion.
· How it works
```
{
  "date": "2026-02-13",
  "context": "I suggested long format for LinkedIn",
  "decision": "reject",
  "reason": "Too formal tone, Bruno prefers casual",
  "tags": ["linkedin", "tone"]
}
```
4 domains: content, tasks, recommendations, digest. Max 30 entries per file (FIFO).
Consolidate patterns into lessons monthly.
Feedback

Granular, JSON
·
Lessons
Curated, prose
·
Decisions
Permanent
Module 04 · Memory
04

MODULE 04 · MEMORY
Build Your AI COO
Semantic Search & Optimization
Memory Search · On-demand Search
The agent doesn't need to load EVERYTHING into memory. It uses semantic search to pull only what it needs:
· Useful commands
memory_search("decision about model for crons")
· Returns ~400 tokens/chunk relevant
memory_get(path="memory/decisions.md", from=15, lines=10)
· Pull specific excerpt
openclaw memory index --all
· Index all memory files
·Token Optimization
Session initialization rule: load ONLY SOUL.md, USER.md, IDENTITY.md and memory/YYYY-MM-DD.md. Use memory_search() on demand for the rest. Reduces from 50KB · 8KB per session (~80% savings).
Periodic Consolidation
Every 15 days, the agent does a complete review:
1. Read all recent daily notes
2. Identify lessons and decisions that escaped
3. Update topic files with what's worth keeping
4. Update MEMORY.md with distilled learnings
5. Delete expired tactical notes (>30 days)
Think of it as a human reviewing their diary and updating their mental model. Daily notes are raw records. MEMORY.md is curated wisdom.
· Module Checkpoint
· memory/ folder created with 5 topic files
· MEMORY.md as index in the root
· Automatic daily notes configured
· Compaction configured (160k context, 30k reserve)

· Extraction rule before compaction in AGENTS.md
· HEARTBEAT.md with periodic checklist
· Feedback loop configured (at least 1 domain)
"If you don't extract before compacting, you lose 80% of the value."
· Lesson from Day 4
·
NEXT MODULE
Module 5 · Integrations & Crons: Connecting to the Real World